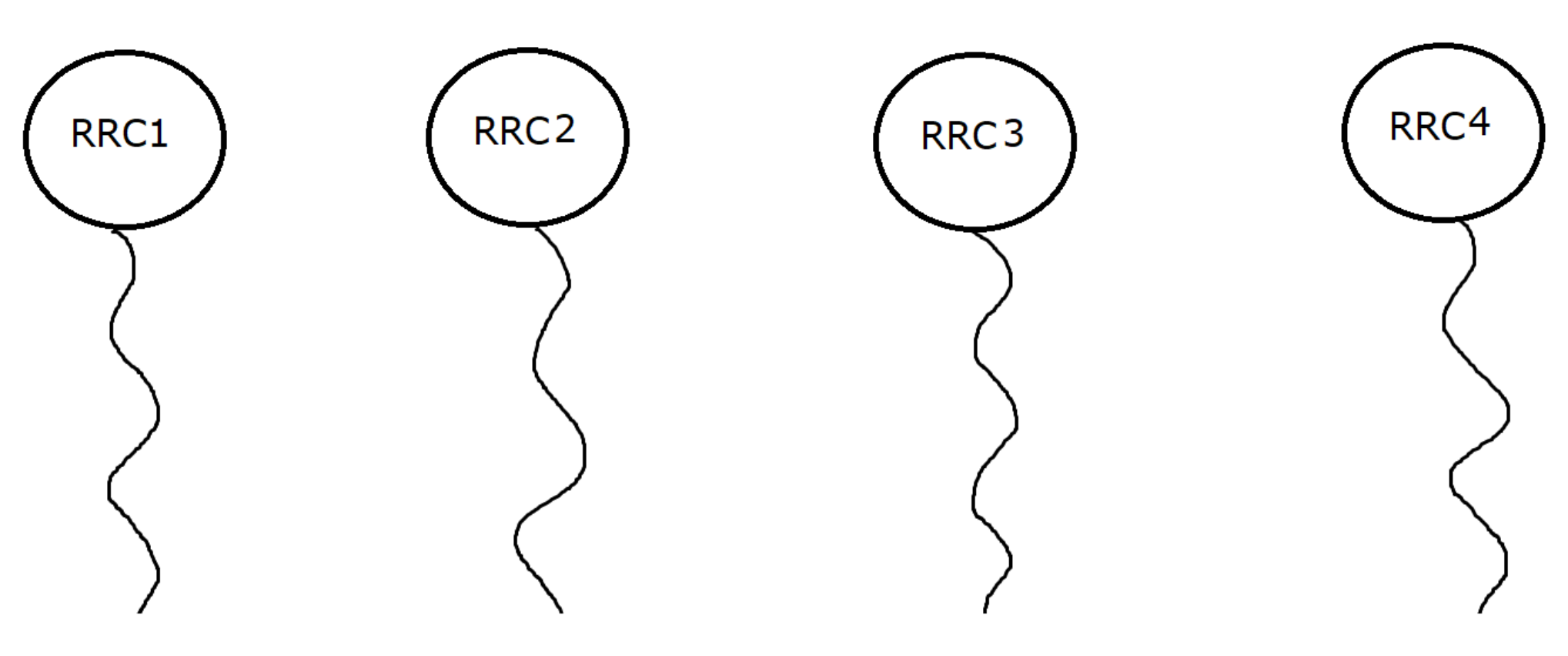
What is a Thread ?



* A thread is the basic unit of CPU which **can run concurrently with another thread within the same process.**

* It is light weight process because a thread will utilize the same memory address, resourses and so on.

* A thread is well known for **"Independent Exceution".**

* The advantage of thread to provide **fast execution.**

* In Java, whenever we define a **main method** then internally one thread is provided by the Operating System, the name of this thread is **"main thread" and it is running under a particular group i.e "main group"**

* The main purpose of this **main thread** to execute the **entire main method**.

WAP to show that main is a thread ?

In java there is a predefined class called Thread available in java.lang package, this class contains a predefined static factory method currentThread() which will provide currently executing Thread Object.

public static native Thread currentThread();

Thread t = Thread.currentThread(); //static Factory Method

Thread class has provided predefined non static method getName() to get the name of the current Thread.
public final String getName();

MainThread.java

```
package com.ravi.basic;

public class MainThread
{
    public static void main(String[] args)
    {
        Thread t = Thread.currentThread();
        IO.println("Current thread name is :"+t.getName());

                       //OR [using Method Chaining]
        String name = Thread.currentThread().getName();
        IO.println("Running thread name is :"+name);
    }
}
```

Note : A single thread i.e. main thread always executes the program **line by line** which is known as Synchronous execution.

How to create our own user defined Thread in java :

We can create user defined Thread by using the following two packages in java :

1) By using java.lang package (JDK 1.0)
2) By using java.util.concurrent sub package (JDK 1.5)

Creating Thread by java.lang Package :

We can create Thred by using java.lang package by using one of the following two ways :

1) By extends java.lang.Thread class
2) By implementing java.lang.Runnable interface

```
public class UserThread implements Runnable
{
}

public class CustomThread extends Thread
{
}
```

```
@FunctionalInterface
public interface Runnable
{
   void run();
}
public class Thread implements Runnable
{
    @Override
    public void run()
    {
    }
   start();
   setName(String name)
   getName()
   setPriority(int newPriority)
   getPriority()
   currentThread()
   isAlive()
   sleep(long ms)
   yield()
   interrupt()
   isInterrupted()
   setDaemon()
   isDaemon()
}
```

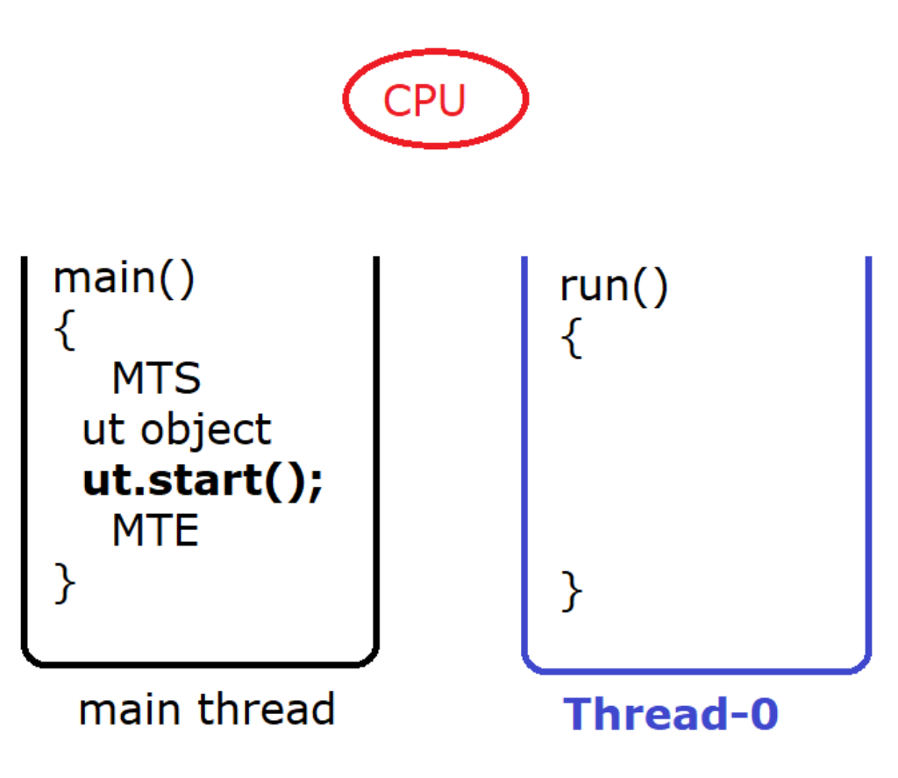
Creating a custom thread by using extend Thread class approach :

```
class UserThread extends Thread
{
     @Override
     public void run()
     {
        //The task which we want to perform through our custom thread
     }
}

public class CustomThread
{
     public static void main(String[] args)
     {
         IO.println("Main thread started.");

         UserThread ut = new UserThread();
         ut.start();

        IO.println("Main thread ended.");
     }
}
```



public void start()
----
* It is a predefined non static method of Thread class.

* Whenever we call start() method, It internally performs two tasks.

  a) It will make a request to the Operating System to assign a **new thread for concurrent execution.**

  b) It will implicitly call the run() method on the same object (current object)